

# Cryptnox CLI Manual

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# 1 Cryptnox CLI Overview

`cryptnox-cli` is a command-line interface for managing **Cryptnox Smart cards**, enabling secure seed initialization and cryptographic signing for **Bitcoin** and **Ethereum**.

## 1.1 Supported Hardware

- **Cryptnox Smart cards**
- **Standard PC/SC Smart card Readers**: either USB NFC reader or a USB smart card reader

Get your card and readers here: shop.cryptnox.com

## 1.2 Installation

> **Note**
>
> This is only a minimal setup. Additional packages may be required depending on your operating system.

### 1.2.1 From PyPI

```
pip install cryptnox-cli
```

### 1.2.2 From source

```
git clone https://github.com/cryptnox/cryptnox-cli.git
cd cryptnox-cli
pip install .
```

This installs the package and makes the `cryptnox` command available (if your Python installation is in your system `PATH`).

## 1.3 Quick Usage Examples

> **Note**
>
> The examples below are only a subset of available commands. The complete list of commands and detailed usage instructions is described in the *Command Line Interface* section.

### 1.3.1 1. Dual initialization

1. Factory reset each card:

`cryptnox reset` → enter PUK → verify reset.

2. Initialize each card:

   `cryptnox init` → (optional) set name/email → set **PIN** (4–9 digits) → set or generate **PUK** → verify init.

3. Run dual seed procedure:

   `cryptnox seed dual` — follow prompts: insert Card A (enter PIN), swap to Card B (enter PIN), swap back as requested.

### 1.3.2 2. Sign and send a Bitcoin transaction

1. Create or obtain a raw unsigned transaction externally.
2. Run the signing & send command:

   ```
   cryptnox btc send <recipient_address> <amount> [-f <fees>]
   ```

### 1.3.3 3. Change PIN code

1. Run command: `cryptnox change_pin`
2. Enter current PIN → enter new PIN → verify change.
3. Check with `cryptnox info` using new PIN (BTC & ETH accounts displayed).

### 1.3.4 4. Get extended public key (xpub)

1. Run command: `cryptnox get_xpub`
2. Enter **PIN** → enter **PUK**
3. The card returns the **xpub**

## 1.4 License

cryptnox-cli is dual-licensed:

- **LGPL-3.0** for open-source projects and proprietary projects that comply with LGPL requirements
- **Commercial license** for projects that require a proprietary license without LGPL obligations (see COMMERCIAL.md for details)

For commercial inquiries, contact: contact@cryptnox.com

# 2 Command Line Interface

The `cryptnox` command provides a comprehensive CLI for managing Cryptnox cards and performing cryptocurrency operations.

## 2.1 Installation

After installing the cryptnox-cli package, the `cryptnox` command becomes available in your system:

```
pip install cryptnox-cli
```

## 2.2 Basic Usage

```
cryptnox [OPTIONS] COMMAND [ARGS]...
```

## 2.3 Global Options

**-v**, **--version**
: Show the version and exit.

**--verbose**
: Turn on logging for detailed output.

**-s**, **--serial** SERIAL
: Serial number of the card to be used for the command.

**--port** PORT
: Define port to enable remote feature.

## 2.4 Commands Overview

### 2.4.1 Card Management Commands

#### list

List all connected Cryptnox cards.

```
cryptnox list
```

#### init

Initialize a Cryptnox card with owner information and PIN/PUK codes.

```
cryptnox init [OPTIONS]
```

**Options:**

- `-e, --easy_mode`: Initialize card in easy mode (sets PIN and PUK to all zeros)

#### reset

Reset the card to factory defaults.

```
cryptnox reset
```

**Warning:** This will erase all data on the card.

#### info

Display default accounts information for the card.

```
cryptnox info
```

**Shows:**

- Card serial number and type
- Ethereum and Bitcoin addresses
- Public keys
- Derivation information

#### cert

Retrieve and display the manufacturer certificate from the card in a human-readable format.

```
cryptnox cert
```

**Features:**

- Retrieves the full manufacturer certificate from the card
- Displays certificate details (Issuer, Subject, Validity, Public Key, Signature)
- Human-readable format similar to `openssl x509 -text`

### 2.4.2 Seed Management Commands

#### seed chip

Generate new root key directly in the card's secure chip.

```
cryptnox seed chip
```

**Note:** The seed never leaves the card and cannot be backed up.

#### seed dual

Generate the same seed on two cards for redundancy.

```
cryptnox seed dual
```

**Requirements:** Two initialized Cryptnox cards

**Process:**

1. Generate seed on first card
2. Swap cards
3. Load same seed on second card
4. Results in two cards with identical keys

#### seed recover

Recover a wallet from an existing BIP39 mnemonic phrase (12 or 24 words).

```
cryptnox seed recover
```

**Input Required:**

- Existing BIP39 mnemonic (12 or 24 words)
- Optional: BIP39 passphrase (13th/25th word)

**Use Cases:**

- Restoring existing wallet
- Migrating wallet from another device
- Recovering from backup mnemonic

**Important:** If the original wallet used a BIP39 passphrase, you **must** provide the same passphrase during recovery.

#### seed upload

Generate new random seed, upload to card, and display BIP39 mnemonic for backup.

```
cryptnox seed upload
```

**Features:**

- Generates new 32-byte random seed using card's hardware RNG
- Converts to BIP39 mnemonic (12 or 24 words)
- Optional: Add BIP39 passphrase (13th/25th word) for extra security
- Displays mnemonic for manual backup

**Output:** The generated mnemonic phrase (save it securely!)

**Important:** If you use a BIP39 passphrase, you **must** remember it. It cannot be recovered.

### 2.4.3 Security Commands

#### change_pin

Change the PIN code of the card.

```
cryptnox change_pin
```

**Requirements:** Current PIN code

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#### change_puk

Change the PUK (PIN Unblocking Key) code of the card.

```
cryptnox change_puk
```

**Requirements:** Current PUK code

#### unlock_pin

Unlock a card with blocked PIN using the PUK code and set a new PIN.

```
cryptnox unlock_pin
```

**Requirements:** PUK code

**Note:** PIN becomes blocked after multiple failed attempts.

### 2.4.4 Card Configuration

#### card_conf

Show or modify card configuration settings.

```
cryptnox card_conf [KEY] [VALUE]
```

**Available Settings:**

- `pinless`: Enable/disable PIN-less path
- `pin`: Enable/disable PIN authentication

**Values:** `yes` or `no`

**Examples:**

```
# Show current configuration
cryptnox card_conf

# Enable PIN-less path
cryptnox card_conf pinless yes

# Disable PIN-less path
cryptnox card_conf pinless no
```

### 2.4.5 Bitcoin Commands

#### btc send

Send Bitcoin to an address.

```
cryptnox btc send ADDRESS AMOUNT [OPTIONS]
```

**Arguments:**

- `ADDRESS`: Bitcoin address (P2PKH, P2SH, or Bech32)
- `AMOUNT`: Amount in BTC

**Options:**

- `-n, --network {mainnet,testnet}`: Network to use
- `-f, --fees SATOSHIS`: Transaction fees in satoshis per byte

**Example:**

```
cryptnox btc send 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa 0.001 -n mainnet -f 10
```

#### btc config

View or modify Bitcoin configuration.

```
cryptnox btc config [KEY] [VALUE]
```

**Settings:**

- `network`: mainnet or testnet
- `derivation`: Key derivation method

### 2.4.6 Ethereum Commands

#### eth send

Send Ether (ETH) to an address.

```
cryptnox eth send ADDRESS AMOUNT [OPTIONS]
```

**Arguments:**

- `ADDRESS`: Ethereum address (0x...)
- `AMOUNT`: Amount in ETH

**Options:**

- `-n, --network {mainnet,sepolia,goerli}`: Network to use
- `--price GWEI`: Gas price in Gwei
- `--limit GAS`: Gas limit

**Example:**

```
cryptnox eth send 0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb 0.1 -n mainnet --
↪price 20
```

#### eth config

View or modify Ethereum configuration.

```
cryptnox eth config [KEY] [VALUE]
```

**Settings:**

- `network`: mainnet, sepolia, or other networks
- `derivation`: Key derivation method

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- `api_key`: Etherscan/Infura API key
- `endpoint`: RPC endpoint

### 2.4.7 ERC Token Commands

#### eth erc20 init

Initialize ERC-20 token contract for transactions.

```
cryptnox eth erc20 init CONTRACT_ADDRESS [OPTIONS]
```

#### eth erc20 info

Display information about configured ERC-20 tokens.

```
cryptnox eth erc20 info
```

#### eth erc20 send

Send ERC-20 tokens.

```
cryptnox eth erc20 send TOKEN_ADDRESS RECIPIENT_ADDRESS AMOUNT [OPTIONS]
```

#### transfer

Transfer ERC-20 or ERC-721 tokens.

```
cryptnox transfer ADDRESS AMOUNT [OPTIONS]
```

**Options:**

- `--price GWEI`: Gas price
- `--limit GAS`: Gas limit

### 2.4.8 Configuration Commands

#### config

List or modify blockchain configurations.

```
cryptnox config [SECTION] [KEY] [VALUE]
```

**Examples:**

```
# Show all configuration
cryptnox config

# Show Ethereum configuration
cryptnox config eth

# Set Ethereum network
cryptnox config eth network mainnet
```

```
# Set Bitcoin derivation
cryptnox config btc derivation DERIVE
```

### 2.4.9 History and Information

#### history

List performed signatures and transactions.

```
cryptnox history [PAGE]
```

**Arguments:**

- `PAGE`: Page number to display (default: 1)

**Note:** Shows up to 148 entries, 25 per page.

### 2.4.10 Advanced Commands

#### get_xpub

Get extended public key (xpub) for hierarchical deterministic wallets.

```
cryptnox get_xpub [OPTIONS]
```

**Options:**

- Derivation path
- Key type

#### get_clearpubkey

Get clear (uncompressed) public key from the card.

```
cryptnox get_clearpubkey [OPTIONS]
```

#### decrypt

Decrypt data using the card's private key.

```
cryptnox decrypt [OPTIONS]
```

### 2.4.11 User Key Management

#### user_key list

List all configured user keys for authentication.

```
cryptnox user_key list
```

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#### user_key add

Add a new user key for authentication (PIV card, Windows Hello).

```
cryptnox user_key add TYPE [DESCRIPTION]
```

**Available Types:**

- `piv`: PIV-compatible smart card
- `hello`: Windows Hello (biometric)

**Example:**

```
cryptnox user_key add piv "My PIV Key"
```

#### user_key delete

Delete a user key.

```
cryptnox user_key delete TYPE
```

### 2.4.12 Server Mode

#### server

Start a server or establish connection to a remote server.

```
cryptnox server [OPTIONS]
```

**Options:**

- `--port PORT`: Server port (default: 5050)
- `--host HOST`: Server host (default: 0.0.0.0)

**Use Case:** Remote card access over network

## 2.5 Interactive Mode

Launch interactive CLI mode:

```
cryptnox
```

**In interactive mode, you can:**

- Execute commands without repeating `cryptnox`
- Use `use` command to switch between multiple cards
- Use `exit` to quit

**Example Session:**

```
$ cryptnox
Cryptnox CLI 1.0.5

> list
```

```
[Shows available cards]

> info
[Shows card information]

> exit
```

## 2.6 BIP39 Passphrase Support

The `seed recover` and `seed upload` commands support BIP39 passphrases (also known as the 13th/25th word).

### 2.6.1 What is a BIP39 Passphrase?

**A BIP39 passphrase is an optional additional word that enhances security:**

- Acts as a “second factor” for your seed
- Creates a completely different wallet from the same mnemonic
- Must be remembered separately (not stored with the mnemonic)
- Cannot be recovered if forgotten

### 2.6.2 Using BIP39 Passphrase

**During seed upload (new wallet):**

```
cryptnox seed upload
```

The command will prompt:

```
Do you want to use a BIP39 passphrase? [y/N]
```

**If you choose `yes`:**

- Enter your desired passphrase
- Confirm the passphrase
- **Remember it!** You’ll need it for recovery

**During seed recover (existing wallet):**

```
cryptnox seed recover
```

If your wallet was created with a passphrase, you **must** provide the same passphrase when recovering.

**Important Notes:**

- **!** If you forget your passphrase, your funds are **permanently lost**
- ✓ Same mnemonic + different passphrase = completely different wallet
- ✓ Passphrase can be any UTF-8 string (including spaces and special characters)

## 2.7 Exit Codes

The CLI returns the following exit codes:

- `0`: Success
- `-1`: Error occurred
- `-2`: Card error or card not found

## 2.8 Error Handling

**When errors occur:**

- Detailed error messages are displayed
- Errors are logged to: `~/.local/share/cryptnox-cli/error.log` (Linux/macOS) or `%LOCALAPPDATA%\\cryptnox\\cryptnox-cli\\error.log` (Windows)
- You can report errors to help improve the application

## 2.9 Examples

**Initialize a new card:**

```
cryptnox init
```

**Generate and upload a seed:**

```
cryptnox seed upload
```

**Recover from existing mnemonic:**

```
cryptnox seed recover
```

**Send Bitcoin:**

```
cryptnox btc send 1A1zP1eP5QGefi2DMPTfTL5SLmv7DivfNa 0.001 -n testnet
```

**Send Ethereum:**

```
cryptnox eth send 0x742d35Cc6634C0532925a3b844Bc9e7595f0bEb 0.1 -n mainnet
```

**View card info:**

```
cryptnox info
```

**Change PIN:**

```
cryptnox change_pin
```

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## 2.10 See Also

- *Cryptnox CLI Overview* - Overview of Cryptnox CLI
- `cryptnox_cli.command.seed` - Seed command implementation details
- *cryptnox_cli.command package* - All command implementations
- *API Reference* - Complete API Reference

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# 3 API Reference

## 3.1 cryptnox_cli package (Cryptnox CLI)

### 3.1.1 Subpackages

#### cryptnox_cli.command package

#### Subpackages

#### cryptnox_cli.command.card package

#### Submodules

#### cryptnox_cli.command.card.info module

#### cryptnox_cli.command.card.initialize module

#### Module contents

The `cryptnox_cli.command.card` package provides card management functionality and exports:

**class** `cryptnox_cli.command.card.`**Info**

Command class for displaying card information. See `cryptnox_cli.command.card.info.Info` for details.

**class** `cryptnox_cli.command.card.`**Initialize**

Command class for initializing cards. See `cryptnox_cli.command.card.initialize.Initialize` for details.

#### cryptnox_cli.command.erc_token package

#### Submodules

#### cryptnox_cli.command.erc_token.contract module

#### cryptnox_cli.command.erc_token.info module

#### cryptnox_cli.command.erc_token.initialize module

#### Module contents

The `cryptnox_cli.command.erc_token` package provides ERC token management and exports:

**class** `cryptnox_cli.command.erc_token.`**Info**

Command class for displaying ERC token information. See `cryptnox_cli.command.erc_token.info.Info` for details.

**class** `cryptnox_cli.command.erc_token.`**Initialize**

Command class for initializing ERC tokens. See `cryptnox_cli.command.erc_token.initialize.Initialize` for details.

### cryptnox_cli.command.helper package

#### Submodules

### cryptnox_cli.command.helper.cards module

### cryptnox_cli.command.helper.config module

### cryptnox_cli.command.helper.helper_methods module

### cryptnox_cli.command.helper.masked_input module

Masked entry of PINs, PUKs and passphrases: one mask character is echoed per character entered, on Windows consoles and POSIX terminals alike.

`cryptnox_cli.command.helper.masked_input.`**`getpass`**(*prompt='Password: '*, *mask='*'*)

Cross-platform getpass that raises KeyboardInterrupt on Ctrl+C.

### cryptnox_cli.command.helper.notification module

### cryptnox_cli.command.helper.security module

### cryptnox_cli.command.helper.ui module

#### Module contents

The `cryptnox_cli.command.helper` package provides utility functions and helper modules for command implementations. This package contains shared functionality used across multiple commands, including user interface components, security helpers, and configuration management.

### cryptnox_cli.command.options package

#### Submodules

### cryptnox_cli.command.options.common module

### cryptnox_cli.command.options.eth module

### cryptnox_cli.command.options.options module

#### Module contents

The `cryptnox_cli.command.options` package provides command-line argument parsing and exports:

`cryptnox_cli.command.options.`**`add`**(*subparsers*, *command_class*)

Add command-line options for a command class. See `cryptnox_cli.command.options.options.add()` for details.

### cryptnox_cli.command.user_keys package

#### Subpackages

### cryptnox_cli.command.user_keys.hello package

#### Submodules

### cryptnox_cli.command.user_keys.hello.exceptions module

### cryptnox_cli.command.user_keys.hello.hello module

### cryptnox_cli.command.user_keys.hello.windows_hello module

#### Module contents

### cryptnox_cli.command.user_keys.piv package

#### Submodules

### cryptnox_cli.command.user_keys.piv.piv module

### cryptnox_cli.command.user_keys.piv.piv_card module

#### Module contents

#### Submodules

### cryptnox_cli.command.user_keys.authentication module

### cryptnox_cli.command.user_keys.user_key_base module

Module containing base class for user key.

**exception** cryptnox_cli.command.user_keys.user_key_base.**UserKeyException**

Bases: Exception

Base exception for all exceptions for UserKey

**exception** cryptnox_cli.command.user_keys.user_key_base.**ExitException**

Bases: UserKeyException

The user has exited the request

**exception** cryptnox_cli.command.user_keys.user_key_base.**NotFoundException**

Bases: UserKeyException

The user key hasn't been found

**exception** cryptnox_cli.command.user_keys.user_key_base.**NotSupportedException**

Bases: UserKeyException

The user key is not supported

**exception** cryptnox_cli.command.user_keys.user_key_base.**ProcessingException**

Bases: UserKeyException

There was an issue in processing the user key

**class** cryptnox_cli.command.user_keys.user_key_base.**UserKey**(*target: str = ''*)

Bases: object

Base class on top of which other user key classes should be implemented

**priority: int = 0**

**__init__**(*target: str = ''*)

**abstractmethod delete**() → None

Delete the key from the user key provider

**abstract property description: str**

Description of the user key service :rtype: str

**Type**
return

**abstract property name: str**

Name of the user key service :rtype: str

**Type**
return

**abstract property public_key: bytes**

Public key of the user key service :rtype: bytes

**Type**
return

**abstractmethod sign**(*message: bytes*) → bytes

Get signature of message from user key service

**Parameters**
**message** (bytes) – Message to sign

**Returns**
Signature of the given message

**Return type**
bytes

**abstract property slot_index: cryptnox_sdk_py.SlotIndex**

Compatible slot index on the card :rtype: cryptnox_sdk_py.SlotIndex

**Type**
return

**property custom_bit: int**

Custom bit to use for checking if method is enabled, -1 for not used :rtype: int

**Type**
return

**added**(*card*)

**classmethod enabled**(*card: cryptnox_sdk_py.Card*) → bool

### Module contents

The `cryptnox_cli.command.user_keys` package provides authentication functionality and exports:

`cryptnox_cli.command.user_keys.`**`add`**`(`*`data`*`)`

Add a new authentication method. See `cryptnox_cli.command.user_keys.authentication.add()` for details.

`cryptnox_cli.command.user_keys.`**`authenticate`**`(`*`data`*`)`

Authenticate using a configured method. See `cryptnox_cli.command.user_keys.authentication.authenticate()` for details.

`cryptnox_cli.command.user_keys.`**`delete`**`(`*`data`*`)`

Delete an authentication method. See `cryptnox_cli.command.user_keys.authentication.delete()` for details.

`cryptnox_cli.command.user_keys.`**`get`**`(`*`data`*`)`

Get information about authentication methods. See `cryptnox_cli.command.user_keys.authentication.get()` for details.

### Submodules

**cryptnox_cli.command.btc module**

**cryptnox_cli.command.card_configuration module**

**cryptnox_cli.command.cards module**

**cryptnox_cli.command.change_pin module**

**cryptnox_cli.command.change_puk module**

**cryptnox_cli.command.command module**

**cryptnox_cli.command.config module**

**cryptnox_cli.command.decrypt module**

**cryptnox_cli.command.eth module**

**cryptnox_cli.command.factory module**

**cryptnox_cli.command.get_clearpubkey module**

**cryptnox_cli.command.get_xpub module**

**cryptnox_cli.command.history module**

**cryptnox_cli.command.info module**

**cryptnox_cli.command.initialize module**

**cryptnox_cli.command.manufacturer_certificate module**

**cryptnox_cli.command.reset module**

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### cryptnox_cli.command.seed module

### cryptnox_cli.command.server module

### cryptnox_cli.command.transfer module

### cryptnox_cli.command.unknown module

### cryptnox_cli.command.unlock_pin module

### cryptnox_cli.command.user_key module

### Module contents

Module to work with commands given by the user in a command line

The `cryptnox_cli.command` package provides command-line interface commands for interacting with Cryptnox cards. This package contains all command implementations that can be executed from the CLI.

## cryptnox_cli.lib package

### Subpackages

### cryptnox_cli.lib.cryptos package

### Subpackages

### cryptnox_cli.lib.cryptos.coins package

### Submodules

### cryptnox_cli.lib.cryptos.coins.base module

Base classes and shared logic for UTXO-based coin implementations.

**class** cryptnox_cli.lib.cryptos.coins.base.**BaseCoin**(*testnet=False*, ***kwargs*)

Bases: `object`

Base implementation of crypto coin class All child coins must follow same pattern.

**coin_symbol = None**

**display_name = None**

**enabled = True**

**segwit_supported = None**

**magicbyte = None**

**script_magicbyte = None**

**segwit_hrp = None**

**client_kwargs = {'host': None, 'loop': None, 'max_servers': 5, 'port': 50001, 'server_file': 'bitcoin.json', 'servers': (), 'timeout': 15}**

**testnet_overrides = {}**

**hashcode = 1**

**hd_path = 0**

**wif_prefix = 128**

**wif_script_types = {'p2pkh': 0, 'p2sh': 5, 'p2wpkh': 1, 'p2wpkh-p2sh': 2, 'p2wsh': 6, 'p2wsh-p2sh': 7}**

**xprv_headers = {'p2pkh': 76066276, 'p2wpkh': 78791436, 'p2wpkh-p2sh': 77428856, 'p2wsh': 44726937, 'p2wsh-p2sh': 43364357}**

**xpub_headers = {'p2pkh': 76067358, 'p2wpkh': 78792518, 'p2wpkh-p2sh': 77429938, 'p2wsh': 44728019, 'p2wsh-p2sh': 43365439}**

**electrum_xprv_headers = {'p2pkh': 76066276, 'p2wpkh': 78791436, 'p2wpkh-p2sh': 77428856, 'p2wsh': 44726937, 'p2wsh-p2sh': 43364357}**

**electrum_xpub_headers = {'p2pkh': 76067358, 'p2wpkh': 78792518, 'p2wpkh-p2sh': 77429938, 'p2wsh': 44728019, 'p2wsh-p2sh': 43365439}**

**__init__**(*testnet=False*, ***kwargs*)

**is_testnet = False**

**address_prefixes = ()**

**secondary_hashcode = None**

**property rpc_client**
Connect to remove server

**unspent**(**addrs*)
Get unspent transactions for addresses

**history**(**addrs*, ***kwargs*)
Get transaction history for addresses

**fetchtx**(*tx*)
Fetch a tx from the blockchain

**txinputs**(*tx*)
Fetch inputs of a transaction on the blockchain

**pushtx**(*tx*)
Push/ Broadcast a transaction to the blockchain

**privtopub**(*privkey*)
Get public key from private key

**pubtoaddr**(*pubkey*)
Get address from a pubic key

**privtoaddr**(*privkey*)
Get address from a private key

**electrum_address**(*masterkey*, *n*, *for_change=0*)

For old electrum seeds

**is_address**(*addr*)

Check if addr is a valid address for this chain

**is_p2sh**(*addr*)

Check if addr is a a pay to script address

**output_script_to_address**(*script*)

Convert an output script to an address

**scripttoaddr**(*script*)

Convert an input public key hash to an address

**p2sh_scriptaddr**(*script*)

Convert an output p2sh script to an address

**addrtoscript**(*addr*)

Convert an output address to a script

**pubtop2w**(*pub*)

Convert a public key to a pay to witness public key hash address (P2WPKH, required for segwit)

**privtop2w**(*priv*)

Convert a private key to a pay to witness public key hash address

**hash_to_segwit_addr**(*hash*)

Convert a hash to the new segwit address format outlined in BIP-0173

**privtosegwit**(*privkey*)

Convert a private key to the new segwit address format outlined in BIP01743

**pubtosegwit**(*pubkey*)

Convert a public key to the new segwit address format outlined in BIP01743

**script_to_p2wsh**(*script*)

Convert a script to the new segwit address format outlined in BIP01743

**mk_multsig_address**(**args*)

**Parameters**

**args** – List of public keys to used to create multisig and M, the number of signatures required to spend

**Returns**

multisig script

**is_segwit**(*priv*, *addr*)

Check if addr was generated from priv using segwit script

**sign**(*txobj*, *i*, *priv*)

Sign a transaction input with index using a private key

**signall**(*txobj*, *priv*)

Sign all inputs to a transaction using a private key

**multisign**(*tx*, *i*, *script*, *pk*)

**mktx**(**args*)

[in0, in1...],[out0, out1...] or in0, in1 ... out0 out1 ...

Make an unsigned transaction from inputs and outputs. Change is not automatically included so any difference in value between inputs and outputs will be given as a miner's fee (transactions with too high fees will normally be blocked by the explorers)

For Bitcoin Cash and other hard forks using SIGHASH_FORKID, ins must be a list of dicts with each containing the outpoint and value of the input.

Inputs originally received with segwit must be a dict in the format: {'outpoint': "txhash:index", value:0, "segwit": True}

For other transactions, inputs can be dicts containing only outpoints or strings in the outpoint format. Outpoint format: txhash:index

**mksend**(**args*, *segwit=False*)

[in0, in1...],[out0, out1...] or in0, in1 ... out0 out1 ...

Make an unsigned transaction from inputs, outputs change address and fee. A change output will be added with change sent to the change address.

For Bitcoin Cash and other hard forks using SIGHASH_FORKID and segwit, ins must be a list of dicts with each containing the outpoint and value of the input.

For other transactions, inputs can be dicts containing only outpoints or strings in the outpoint format. Outpoint format: txhash:index

**preparesignedtx**(*privkey*, *to*, *value*, *fee=10000*, *change_addr=None*, *segwit=False*, *addr=None*)

Prepare a tx with a specific amount from address belonging to private key to another address, returning change to the from address or change address, if set. Requires private key, target address, value and optionally the change address and fee segwit paramater specifies that the inputs belong to a segwit address addr, if provided, will explicity set the from address, overriding the auto-detection of the address from the private key.It will also be used, along with the privkey, to automatically detect a segwit transaction for coins which support segwit, overriding the segwit kw

**send**(*privkey*, *to*, *value*, *fee=10000*, *change_addr=None*, *segwit=False*, *addr=None*)

Send a specific amount from address belonging to private key to another address, returning change to the from address or change address, if set. Requires private key, target address, value and optionally the change address and fee segwit paramater specifies that the inputs belong to a segwit address addr, if provided, will explicity set the from address, overriding the auto-detection of the address from the private key.It will also be used, along with the privkey, to automatically detect a segwit transaction for coins which support segwit, overriding the segwit kw

**preparesignedmultitx**(*privkey*, **args*, *change_addr=None*, *segwit=False*, *addr=None*)

Prepare transaction with multiple outputs, with change sent back to from addrss

Requires private key, address:value pairs and optionally the change address and fee segwit paramater specifies that the inputs belong to a segwit address addr, if provided, will explicity set the from address, overriding the auto-detection of the address from the private key.It will also be used, along with the privkey to automatically detect a segwit transaction for coins which support segwit, overriding the segwit kw

**sendmultitx**(*privkey*, **args*, *change_addr=None*, *segwit=False*, *addr=None*)

Send transaction with multiple outputs, with change sent back to from addrss Requires private key, address:value pairs and optionally the change address and fee segwit paramater specifies that the inputs belong to a segwit address addr, if provided, will explicity set the from address, overriding the auto-detection of the address from the private key.It will also be used, along with the privkey to automatically detect a segwit transaction for coins which support segwit, overriding the segwit kw

**preparetx**(*frm*, *to*, *value*, *fee*, *change_addr=None*, *segwit=False*)

Prepare a transaction using from and to addresses, value and a fee, with change sent back to from address

**preparemultitx**(*frm*, **args*, *change_addr=None*, *segwit=False*)

Prepare transaction with multiple outputs, with change sent to from address Requires from address, to_address:value pairs and fees

**block_height**(*txhash*)

**current_block_height**()

**block_info**(*height*)

**inspect**(*tx*)

**merkle_prove**(*txhash*)

Prove that information an explorer returns about a transaction in the blockchain is valid. Only run on a tx with at least 1 confirmation.

**encode_privkey**(*privkey*, *formt*, *script_type='p2pkh'*)

**wallet**(*seed*, *passphrase=None*, ***kwargs*)

**watch_wallet**(*xpub*, ***kwargs*)

**p2wpkh_p2sh_wallet**(*seed*, *passphrase=None*, ***kwargs*)

**watch_p2wpkh_p2sh_wallet**(*xpub*, ***kwargs*)

**p2wpkh_wallet**(*seed*, *passphrase=None*, ***kwargs*)

**watch_p2wpkh_wallet**(*xpub*, ***kwargs*)

**electrum_wallet**(*seed*, *passphrase=None*, ***kwargs*)

**watch_electrum_wallet**(*xpub*, ***kwargs*)

**watch_electrum_p2wpkh_wallet**(*xpub*, ***kwargs*)

### cryptnox_cli.lib.cryptos.coins.bitcoin module

Bitcoin-specific constants, scripts, and address helpers.

**class** cryptnox_cli.lib.cryptos.coins.bitcoin.**Bitcoin**(*testnet=False*, ***kwargs*)

Bases: BaseCoin

**coin_symbol = 'BTC'**

**display_name = 'Bitcoin'**

**segwit_supported = True**

**magicbyte = 0**

**script_magicbyte = 5**

**segwit_hrp = 'bc'**

**client_kwargs = {'server_file': 'bitcoin.json'}**

**testnet_overrides = {'client_kwargs': {'server_file': 'bitcoin_testnet.json'}, 'coin_symbol': 'BTCTEST', 'display_name': 'Bitcoin Testnet', 'hd_path': 1, 'magicbyte': 111, 'script_magicbyte': 196, 'segwit_hrp': 'tb', 'wif_prefix': 239, 'xprv_headers': {'p2pkh': 70615956, 'p2wpkh': 70615956, 'p2wpkh-p2sh': 71978536, 'p2wsh': 44726937, 'p2wsh-p2sh': 43364357}, 'xpub_headers': {'p2pkh': 70617039, 'p2wpkh': 70617039, 'p2wpkh-p2sh': 71979618, 'p2wsh': 44728019, 'p2wsh-p2sh': 43365439}}**

### Module contents

Re-exports available coin implementations for convenience.

**class** cryptnox_cli.lib.cryptos.coins.**Bitcoin**(*testnet=False*, ***kwargs*)

Bases: BaseCoin

**coin_symbol = 'BTC'**

**display_name = 'Bitcoin'**

**segwit_supported = True**

**magicbyte = 0**

**script_magicbyte = 5**

**segwit_hrp = 'bc'**

**client_kwargs = {'server_file': 'bitcoin.json'}**

**testnet_overrides = {'client_kwargs': {'server_file': 'bitcoin_testnet.json'}, 'coin_symbol': 'BTCTEST', 'display_name': 'Bitcoin Testnet', 'hd_path': 1, 'magicbyte': 111, 'script_magicbyte': 196, 'segwit_hrp': 'tb', 'wif_prefix': 239, 'xprv_headers': {'p2pkh': 70615956, 'p2wpkh': 70615956, 'p2wpkh-p2sh': 71978536, 'p2wsh': 44726937, 'p2wsh-p2sh': 43364357}, 'xpub_headers': {'p2pkh': 70617039, 'p2wpkh': 70617039, 'p2wpkh-p2sh': 71979618, 'p2wsh': 44728019, 'p2wsh-p2sh': 43365439}}**

The cryptnox_cli.lib.cryptos.coins package provides cryptocurrency coin implementations and exports:

**class** cryptnox_cli.lib.cryptos.coins.**Bitcoin**

Bitcoin cryptocurrency implementation. See cryptnox_cli.lib.cryptos.coins.bitcoin.Bitcoin for details.

### Submodules

#### cryptnox_cli.lib.cryptos.blocks module

Block header encoding/decoding utilities and related primitives.

cryptnox_cli.lib.cryptos.blocks.**serialize_header**(*inp*)

cryptnox_cli.lib.cryptos.blocks.**deserialize_header**(*inp*)

cryptnox_cli.lib.cryptos.blocks.**mk_merkle_proof**(*header*, *hashes*, *index*)

#### cryptnox_cli.lib.cryptos.composite module

High-level composition utilities combining deterministic keys and transactions.

cryptnox_cli.lib.cryptos.composite.**bip32_hdm_script**(**args*)

cryptnox_cli.lib.cryptos.composite.**bip32_hdm_addr**(**args*)

cryptnox_cli.lib.cryptos.composite.**setup_coinvault_tx**(*tx*, *script*)

cryptnox_cli.lib.cryptos.composite.**sign_coinvault_tx**(*tx*, *priv*)

#### cryptnox_cli.lib.cryptos.constants module

Core constants and magic values used across cryptocurrency modules.

#### cryptnox_cli.lib.cryptos.deterministic module

Deterministic key and path derivation utilities (BIP32-like logic).

cryptnox_cli.lib.cryptos.deterministic.**electrum_stretch**(*seed*)

cryptnox_cli.lib.cryptos.deterministic.**electrum_mpk**(*seed*)

cryptnox_cli.lib.cryptos.deterministic.**electrum_privkey**(*seed*, *n*, *for_change=0*)

cryptnox_cli.lib.cryptos.deterministic.**electrum_pubkey**(*masterkey*, *n*, *for_change=0*)

cryptnox_cli.lib.cryptos.deterministic.**electrum_address**(*masterkey*, *n*, *for_change=0*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.deterministic.**crack_electrum_wallet**(*mpk*, *pk*, *n*, *for_change=0*)

cryptnox_cli.lib.cryptos.deterministic.**raw_bip32_ckd**(*rawtuple*, *i*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_serialize**(*rawtuple*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_deserialize**(*data*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**is_xprv**(*text*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**is_xpub**(*text*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**raw_bip32_privtopub**(*rawtuple*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_privtopub**(*data*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_ckd**(*key*, *path*, *prefixes=DEFAULT*, *public=False*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_master_key**(*seed*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_bin_extract_key**(*data*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_extract_key**(*data*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_derive_key**(*key*, *path*, *prefixes=DEFAULT*, ***kwargs*)

cryptnox_cli.lib.cryptos.deterministic.**raw_crack_bip32_privkey**(*parent_pub*, *priv*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**crack_bip32_privkey**(*parent_pub*, *priv*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**coinvault_pub_to_bip32**(**args*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**coinvault_priv_to_bip32**(**args*, *prefixes=DEFAULT*)

cryptnox_cli.lib.cryptos.deterministic.**bip32_descend**(**args*, *prefixes=DEFAULT*)

Descend masterkey and return privkey

cryptnox_cli.lib.cryptos.deterministic.**parse_bip32_path**(*path*)

Takes bip32 path, “m/0’/2H” or “m/0H/1/2H/2/1000000000.pub”, returns list of ints

### cryptnox_cli.lib.cryptos.keystore module

HD wallet structures, serialization, derivation, and key store helpers.

**class** cryptnox_cli.lib.cryptos.keystore.**KeyStore**(*coin*, *addresses=()*)

Bases: object

**__init__**(*coin*, *addresses=()*)

**has_seed**()

**is_watching_only**()

**can_import**()

**get_tx_derivations**(*tx*)

**can_sign**(*tx*)

**class** cryptnox_cli.lib.cryptos.keystore.**Software_KeyStore**(*coin*, *addresses=()*)

Bases: KeyStore

**may_have_password**()

**class** cryptnox_cli.lib.cryptos.keystore.**Imported_KeyStore**(*d*, *coin*)

Bases: Software_KeyStore

**__init__**(*d*, *coin*)

**is_deterministic**()

**can_change_password**()

**get_master_public_key**()

**dump**()

**can_import**()

**check_password**(*password=None*)

**import_privkey**(*sec*, *password=None*)

**delete_imported_key**(*key*)

**get_private_key**(*pubkey*, *password=None*)

**get_pubkey_derivation**(*x_pubkey*)

**update_password**(*old_password*, *new_password*)

**class** cryptnox_cli.lib.cryptos.keystore.**Deterministic_KeyStore**(*d*, *coin*)

Bases: Software_KeyStore

**__init__**(*d*, *coin*)

**is_deterministic**()

**dump**()

**has_seed**()

**is_watching_only**()

**can_change_password**()

**add_seed**(*seed*)

**get_seed**(*password*)

**get_passphrase**(*password*)

**class** cryptnox_cli.lib.cryptos.keystore.**Xpub**

Bases: object

**__init__**()

**get_master_public_key**()

**derive_pubkey**(*for_change*, *n*)

**classmethod get_pubkey_from_xpub**(*xpub*, *sequence*, *bip39_prefixes*)

**class** cryptnox_cli.lib.cryptos.keystore.**BIP32_KeyStore**(*d*, *coin*)

Bases: Deterministic_KeyStore, Xpub

**__init__**(*d*, *coin*)

**format_seed**(*seed*)

**dump**()

**get_master_private_key**(*password=None*)

**check_password**(*password=None*)

**update_password**(*old_password*, *new_password*)

**is_watching_only**()

**add_xprv**(*xprv*)

**add_xpub**(*xpub*, *xtype*, *electrum=False*)

**add_xprv_from_seed**(*bip32_seed*, *xtype*, *derivation*, *electrum=False*)

**get_private_key**(*sequence*, *password*)

**class** cryptnox_cli.lib.cryptos.keystore.**Hardware_KeyStore**(*d*, *coin*)

Bases: KeyStore, Xpub

**max_change_outputs = 1**

**__init__**(*d*, *coin*)

**set_label**(*label*)

**may_have_password**()

**is_deterministic**()

**dump**()

**unpaired**()

A device paired with the wallet was diconnected. This can be called in any thread context.

**paired**()

A device paired with the wallet was (re-)connected. This can be called in any thread context.

**can_export**()

**is_watching_only**()

The wallet is not watching-only; the user will be prompted for pin and passphrase as appropriate when needed.

**can_change_password**()

cryptnox_cli.lib.cryptos.keystore.**bip39_to_seed**(*mnemonic*, *passphrase*)

cryptnox_cli.lib.cryptos.keystore.**bip39_is_checksum_valid**(*mnemonic*)

cryptnox_cli.lib.cryptos.keystore.**from_bip39_seed**(*seed*, *passphrase*, *derivation*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**standard_from_bip39_seed**(*seed*, *passphrase*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**p2wpkh_from_bip39_seed**(*seed*, *passphrase*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**p2wpkh_p2sh_from_bip39_seed**(*seed*, *passphrase*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**xtype_from_derivation**(*derivation*)

Returns the script type to be used for this derivation.

cryptnox_cli.lib.cryptos.keystore.**is_xpubkey**(*x_pubkey*)

cryptnox_cli.lib.cryptos.keystore.**parse_xpubkey**(*x_pubkey*)

cryptnox_cli.lib.cryptos.keystore.**xpubkey_to_address**(*x_pubkey*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**xpubkey_to_pubkey**(*x_pubkey*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**register_keystore**(*hw_type*, *constructor*)

cryptnox_cli.lib.cryptos.keystore.**hardware_keystore**(*d*)

cryptnox_cli.lib.cryptos.keystore.**is_address_list**(*text*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**get_private_keys**(*text*)

cryptnox_cli.lib.cryptos.keystore.**is_private_key_list**(*text*)

cryptnox_cli.lib.cryptos.keystore.**is_private**(*x*)

cryptnox_cli.lib.cryptos.keystore.**is_master_key**(*x*)

cryptnox_cli.lib.cryptos.keystore.**is_private_key**(*x*)

cryptnox_cli.lib.cryptos.keystore.**is_bip32_key**(*x*)

cryptnox_cli.lib.cryptos.keystore.**from_electrum_seed**(*seed*, *passphrase*, *is_p2sh*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**from_private_key_list**(*text*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**from_xpub**(*xpub*, *coin*, *xtype*, *electrum=False*)

cryptnox_cli.lib.cryptos.keystore.**from_xprv**(*xprv*, *coin*)

cryptnox_cli.lib.cryptos.keystore.**from_master_key**(*text*, *coin*)

### cryptnox_cli.lib.cryptos.main module

Core cryptographic and encoding primitives used by higher-level modules.

This module is based on code from pybitcointools: https://github.com/primal100/pybitcointools/blob/master/cryptos/main.py

cryptnox_cli.lib.cryptos.main.**change_curve**(*p*, *n*, *a*, *b*, *gx*, *gy*)

cryptnox_cli.lib.cryptos.main.**getG**()

cryptnox_cli.lib.cryptos.main.**inv**(*a*, *n*)

cryptnox_cli.lib.cryptos.main.**access**(*obj*, *prop*)

cryptnox_cli.lib.cryptos.main.**multiaccess**(*obj*, *prop*)

cryptnox_cli.lib.cryptos.main.**slice**(*obj*, *start=0*, *end=2**200*)

cryptnox_cli.lib.cryptos.main.**count**(*obj*)

cryptnox_cli.lib.cryptos.main.**sum**(*obj*)

cryptnox_cli.lib.cryptos.main.**isinf**(*p*)

cryptnox_cli.lib.cryptos.main.**to_jacobian**(*p*)

cryptnox_cli.lib.cryptos.main.**jacobian_double**(*p*)

cryptnox_cli.lib.cryptos.main.**jacobian_add**(*p*, *q*)

cryptnox_cli.lib.cryptos.main.**from_jacobian**(*p*)

cryptnox_cli.lib.cryptos.main.**jacobian_multiply**(*a*, *n*)

cryptnox_cli.lib.cryptos.main.**fast_multiply**(*a*, *n*)

cryptnox_cli.lib.cryptos.main.**fast_add**(*a*, *b*)

cryptnox_cli.lib.cryptos.main.**get_pubkey_format**(*pub*)

cryptnox_cli.lib.cryptos.main.**encode_pubkey**(*pub*, *formt*)

cryptnox_cli.lib.cryptos.main.**decode_pubkey**(*pub*, *formt=None*)

cryptnox_cli.lib.cryptos.main.**get_privkey_format**(*priv*)

cryptnox_cli.lib.cryptos.main.**encode_privkey**(*priv*, *formt*, *vbyte=128*)

cryptnox_cli.lib.cryptos.main.**decode_privkey**(*priv*, *formt=None*)

cryptnox_cli.lib.cryptos.main.**add_pubkeys**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**add_privkeys**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**mul_privkeys**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**multiply**(*pubkey*, *privkey*)

cryptnox_cli.lib.cryptos.main.**divide**(*pubkey*, *privkey*)

cryptnox_cli.lib.cryptos.main.**compress**(*pubkey*)

cryptnox_cli.lib.cryptos.main.**decompress**(*pubkey*)

cryptnox_cli.lib.cryptos.main.**privkey_to_pubkey**(*privkey*)

cryptnox_cli.lib.cryptos.main.**privtopub**(*privkey*)

cryptnox_cli.lib.cryptos.main.**privkey_to_address**(*priv*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.main.**privtoaddr**(*priv*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.main.**neg_pubkey**(*pubkey*)

cryptnox_cli.lib.cryptos.main.**neg_privkey**(*privkey*)

cryptnox_cli.lib.cryptos.main.**subtract_pubkeys**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**subtract_privkeys**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**bin_hash160**(*string*)

cryptnox_cli.lib.cryptos.main.**hash160**(*string*)

cryptnox_cli.lib.cryptos.main.**hex_to_hash160**(*s_hex*)

cryptnox_cli.lib.cryptos.main.**bin_sha256**(*string*)

cryptnox_cli.lib.cryptos.main.**sha256**(*string*)

cryptnox_cli.lib.cryptos.main.**bin_ripemd160**(*string*)

cryptnox_cli.lib.cryptos.main.**ripemd160**(*string*)

cryptnox_cli.lib.cryptos.main.**bin_dbl_sha256**(*s*)

cryptnox_cli.lib.cryptos.main.**dbl_sha256**(*string*)

cryptnox_cli.lib.cryptos.main.**bin_slowsha**(*string*)

cryptnox_cli.lib.cryptos.main.**slowsha**(*string*)

cryptnox_cli.lib.cryptos.main.**hash_to_int**(*x*)

cryptnox_cli.lib.cryptos.main.**num_to_var_int**(*x*)

cryptnox_cli.lib.cryptos.main.**electrum_sig_hash**(*message*)

cryptnox_cli.lib.cryptos.main.**random_key**()

cryptnox_cli.lib.cryptos.main.**random_electrum_seed**()

cryptnox_cli.lib.cryptos.main.**b58check_to_bin**(*inp*)

cryptnox_cli.lib.cryptos.main.**get_version_byte**(*inp*)

cryptnox_cli.lib.cryptos.main.**hex_to_b58check**(*inp*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.main.**b58check_to_hex**(*inp*)

cryptnox_cli.lib.cryptos.main.**pubkey_to_hash**(*pubkey*)

cryptnox_cli.lib.cryptos.main.**pubkey_to_hash_hex**(*pubkey*)

cryptnox_cli.lib.cryptos.main.**pubkey_to_address**(*pubkey*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.main.**pubtoaddr**(*pubkey*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.main.**is_privkey**(*priv*)

cryptnox_cli.lib.cryptos.main.**is_pubkey**(*pubkey*)

cryptnox_cli.lib.cryptos.main.**encode_sig**(*v*, *r*, *s*)

cryptnox_cli.lib.cryptos.main.**decode_sig**(*sig*)

cryptnox_cli.lib.cryptos.main.**deterministic_generate_k**(*msghash*, *priv*)

cryptnox_cli.lib.cryptos.main.**ecdsa_raw_sign**(*msghash*, *priv*)

cryptnox_cli.lib.cryptos.main.**ecdsa_sign**(*msg*, *priv*, *coin*)

cryptnox_cli.lib.cryptos.main.**ecdsa_raw_verify**(*msghash*, *vrs*, *pub*)

cryptnox_cli.lib.cryptos.main.**ecdsa_verify_addr**(*msg*, *sig*, *addr*, *coin*)

cryptnox_cli.lib.cryptos.main.**ecdsa_verify**(*msg*, *sig*, *pub*, *coin*)

cryptnox_cli.lib.cryptos.main.**ecdsa_raw_recover**(*msghash*, *vrs*)

cryptnox_cli.lib.cryptos.main.**ecdsa_recover**(*msg*, *sig*)

cryptnox_cli.lib.cryptos.main.**add**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**subtract**(*p1*, *p2*)

cryptnox_cli.lib.cryptos.main.**magicbyte_to_prefix**(*magicbyte*)

### cryptnox_cli.lib.cryptos.mnemonic module

Mnemonic phrase generation and seed handling (BIP39-style operations).

This module is based on code from pybtctools: https://github.com/danvergara/pybtctools/blob/master/bitcoin/mnemonic.py

cryptnox_cli.lib.cryptos.mnemonic.**is_CJK**(*c*)

cryptnox_cli.lib.cryptos.mnemonic.**normalize_text**(*seed*)

cryptnox_cli.lib.cryptos.mnemonic.**eint_to_bytes**(*entint*, *entbits*)

cryptnox_cli.lib.cryptos.mnemonic.**mnemonic_int_to_words**(*mint*, *mint_num_words*, *wordlist=wordlist_english*)

cryptnox_cli.lib.cryptos.mnemonic.**entropy_cs**(*entbytes*)

cryptnox_cli.lib.cryptos.mnemonic.**entropy_to_words**(*entbytes*, *wordlist=wordlist_english*)

cryptnox_cli.lib.cryptos.mnemonic.**words_bisect**(*word*, *wordlist=wordlist_english*)

cryptnox_cli.lib.cryptos.mnemonic.**words_split**(*wordstr*, *wordlist=wordlist_english*)

cryptnox_cli.lib.cryptos.mnemonic.**words_to_mnemonic_int**(*words*, *wordlist=wordlist_english*)

cryptnox_cli.lib.cryptos.mnemonic.**words_verify**(*words*, *wordlist=wordlist_english*)

cryptnox_cli.lib.cryptos.mnemonic.**bip39_normalize_passphrase**(*passphrase*)

cryptnox_cli.lib.cryptos.mnemonic.**bip39_is_checksum_valid**(*mnemonic*)

cryptnox_cli.lib.cryptos.mnemonic.**mnemonic_to_seed**(*mnemonic_phrase*, *passphrase=''*, *passphrase_prefix=b'mnemonic'*)

cryptnox_cli.lib.cryptos.mnemonic.**bip39_mnemonic_to_seed**(*mnemonic_phrase*, *passphrase=''*)

cryptnox_cli.lib.cryptos.mnemonic.**electrum_mnemonic_to_seed**(*mnemonic_phrase*, *passphrase=''*)

cryptnox_cli.lib.cryptos.mnemonic.**is_old_seed**(*seed*)

cryptnox_cli.lib.cryptos.mnemonic.**seed_prefix**(*seed_type*)

cryptnox_cli.lib.cryptos.mnemonic.**seed_type**(*x*)

cryptnox_cli.lib.cryptos.mnemonic.**is_seed**(*x*)

cryptnox_cli.lib.cryptos.mnemonic.**words_mine**(*prefix*, *entbits*, *satisfunction*, *wordlist=wordlist_english*, *randombits=random.getrandbits*)

### cryptnox_cli.lib.cryptos.pbkdf2 module

PBKDF2 key derivation function implementation and helpers.

cryptnox_cli.lib.cryptos.pbkdf2.**isunicode**(*s*)

cryptnox_cli.lib.cryptos.pbkdf2.**isbytes**(*s*)

cryptnox_cli.lib.cryptos.pbkdf2.**isinteger**(*n*)

cryptnox_cli.lib.cryptos.pbkdf2.**callable**(*obj*)

cryptnox_cli.lib.cryptos.pbkdf2.**b**(*s*)

cryptnox_cli.lib.cryptos.pbkdf2.**binxor**(*a*, *b*)

cryptnox_cli.lib.cryptos.pbkdf2.**b64encode**(*data*, *chars='+/'*)

cryptnox_cli.lib.cryptos.pbkdf2.**b2a_hex**(*s*)

**class** cryptnox_cli.lib.cryptos.pbkdf2.**PBKDF2**(*passphrase*, *salt*, *iterations=1000*, *digestmodule=SHA1*, *macmodule=HMAC*)

Bases: object

**__init__**(*passphrase*, *salt*, *iterations=1000*, *digestmodule=SHA1*, *macmodule=HMAC*)

**static crypt**(*word*, *salt=None*, *iterations=None*)

PBKDF2-based unix crypt(3) replacement.

The number of iterations specified in the salt overrides the 'iterations' parameter.

The effective hash length is 192 bits.

**read**(*bytes*)

Read the specified number of key bytes.

**hexread**(*octets*)

Read the specified number of octets. Return them as hexadecimal.

Note that len(obj.hexread(n)) == 2*n.

**close**()

Close the stream.

cryptnox_cli.lib.cryptos.pbkdf2.**crypt**(*word*, *salt=None*, *iterations=None*)

PBKDF2-based unix crypt(3) replacement.

The number of iterations specified in the salt overrides the 'iterations' parameter.

The effective hash length is 192 bits.

### cryptnox_cli.lib.cryptos.ripemd module

RIPEMD-160 hash function implementation in pure Python.

**class** cryptnox_cli.lib.cryptos.ripemd.**RIPEMD160**(*arg=None*)

Bases: object

Return a new RIPEMD160 object. An optional string argument may be provided; if present, this string will be automatically hashed.

**__init__**(*arg=None*)

**update**(*arg*)

**digest**()

**hexdigest**()

**copy**()

cryptnox_cli.lib.cryptos.ripemd.**new**(*arg=None*)

Return a new RIPEMD160 object. An optional string argument may be provided; if present, this string will be automatically hashed.

**class** cryptnox_cli.lib.cryptos.ripemd.**RMDContext**

Bases: object

**__init__**()

**copy**()

cryptnox_cli.lib.cryptos.ripemd.**ROL**(*n*, *x*)

cryptnox_cli.lib.cryptos.ripemd.**F0**(*x*, *y*, *z*)

cryptnox_cli.lib.cryptos.ripemd.**F1**(*x*, *y*, *z*)

cryptnox_cli.lib.cryptos.ripemd.**F2**(*x*, *y*, *z*)

cryptnox_cli.lib.cryptos.ripemd.**F3**(*x*, *y*, *z*)

cryptnox_cli.lib.cryptos.ripemd.**F4**(*x*, *y*, *z*)

cryptnox_cli.lib.cryptos.ripemd.**R**(*a*, *b*, *c*, *d*, *e*, *Fj*, *Kj*, *sj*, *rj*, *X*)

cryptnox_cli.lib.cryptos.ripemd.**RMD160Transform**(*state*, *block*)

cryptnox_cli.lib.cryptos.ripemd.**RMD160Update**(*ctx*, *inp*, *inplen*)

cryptnox_cli.lib.cryptos.ripemd.**RMD160Final**(*ctx*)

### cryptnox_cli.lib.cryptos.segwit_addr module

SegWit address encoding/decoding utilities (bech32/bech32m).

cryptnox_cli.lib.cryptos.segwit_addr.**bech32_polymod**(*values*)
Internal function that computes the Bech32 checksum.

cryptnox_cli.lib.cryptos.segwit_addr.**bech32_hrp_expand**(*hrp*)
Expand the HRP into values for checksum computation.

cryptnox_cli.lib.cryptos.segwit_addr.**bech32_verify_checksum**(*hrp*, *data*)
Verify a checksum given HRP and converted data characters.

cryptnox_cli.lib.cryptos.segwit_addr.**bech32_create_checksum**(*hrp*, *data*)
Compute the checksum values given HRP and data.

cryptnox_cli.lib.cryptos.segwit_addr.**bech32_encode**(*hrp*, *data*)
Compute a Bech32 string given HRP and data values.

cryptnox_cli.lib.cryptos.segwit_addr.**bech32_decode**(*bech*)
Validate a Bech32 string, and determine HRP and data.

cryptnox_cli.lib.cryptos.segwit_addr.**convertbits**(*data*, *frombits*, *tobits*, *pad=True*)
General power-of-2 base conversion.

cryptnox_cli.lib.cryptos.segwit_addr.**decode**(*hrp*, *addr*)
Decode a segwit address.

cryptnox_cli.lib.cryptos.segwit_addr.**encode**(*hrp*, *witver*, *witprog*)
Encode a segwit address.

### cryptnox_cli.lib.cryptos.specials module

Low-level helpers, type adapters, and encoding utilities shared across modules.

cryptnox_cli.lib.cryptos.specials.**bin_dbl_sha256**(*s*)

cryptnox_cli.lib.cryptos.specials.**lpad**(*msg*, *symbol*, *length*)

cryptnox_cli.lib.cryptos.specials.**get_code_string**(*base*)

cryptnox_cli.lib.cryptos.specials.**changebase**(*string*, *frm*, *to*, *minlen=0*)

cryptnox_cli.lib.cryptos.specials.**bin_to_b58check**(*inp*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.specials.**bytes_to_hex_string**(*b*)

cryptnox_cli.lib.cryptos.specials.**safe_from_hex**(*s*)

cryptnox_cli.lib.cryptos.specials.**from_int_representation_to_bytes**(*a*)

cryptnox_cli.lib.cryptos.specials.**from_int_to_byte**(*a*)

cryptnox_cli.lib.cryptos.specials.**from_byte_to_int**(*a*)

cryptnox_cli.lib.cryptos.specials.**from_string_to_bytes**(*a*)

cryptnox_cli.lib.cryptos.specials.**safe_hexlify**(*a*)

cryptnox_cli.lib.cryptos.specials.**encode**(*val*, *base*, *minlen=0*)

cryptnox_cli.lib.cryptos.specials.**decode**(*string*, *base*)

cryptnox_cli.lib.cryptos.specials.**random_string**(*x*)

### cryptnox_cli.lib.cryptos.stealth module

Stealth address generation and parsing utilities.

cryptnox_cli.lib.cryptos.stealth.**shared_secret_sender**(*scan_pubkey*, *ephem_privkey*)

cryptnox_cli.lib.cryptos.stealth.**shared_secret_receiver**(*ephem_pubkey*, *scan_privkey*)

cryptnox_cli.lib.cryptos.stealth.**uncover_pay_pubkey_sender**(*scan_pubkey*, *spend_pubkey*, *ephem_privkey*)

cryptnox_cli.lib.cryptos.stealth.**uncover_pay_pubkey_receiver**(*scan_privkey*, *spend_pubkey*, *ephem_pubkey*)

cryptnox_cli.lib.cryptos.stealth.**uncover_pay_privkey**(*scan_privkey*, *spend_privkey*, *ephem_pubkey*)

cryptnox_cli.lib.cryptos.stealth.**pubkeys_to_basic_stealth_address**(*scan_pubkey*, *spend_pubkey*, *magic_byte=42*)

cryptnox_cli.lib.cryptos.stealth.**basic_stealth_address_to_pubkeys**(*stealth_address*)

cryptnox_cli.lib.cryptos.stealth.**mk_stealth_metadata_script**(*ephem_pubkey*, *nonce*)

cryptnox_cli.lib.cryptos.stealth.**mk_stealth_tx_outputs**(*stealth_addr*, *value*, *ephem_privkey*, *nonce*, *network='btc'*)

cryptnox_cli.lib.cryptos.stealth.**ephem_pubkey_from_tx_script**(*stealth_tx_script*)

### cryptnox_cli.lib.cryptos.transaction module

Transaction building, serialization, signing, and parsing utilities.

cryptnox_cli.lib.cryptos.transaction.**json_is_base**(*obj*, *base*)

cryptnox_cli.lib.cryptos.transaction.**json_changebase**(*obj*, *changer*)

cryptnox_cli.lib.cryptos.transaction.**encode_1_byte**(*val*)

cryptnox_cli.lib.cryptos.transaction.**encode_4_bytes**(*val*)

cryptnox_cli.lib.cryptos.transaction.**encode_8_bytes**(*val*)

cryptnox_cli.lib.cryptos.transaction.**list_to_bytes**(*vals*)

cryptnox_cli.lib.cryptos.transaction.**dbl_sha256_list**(*vals*)

cryptnox_cli.lib.cryptos.transaction.**is_segwit**(*tx*)

cryptnox_cli.lib.cryptos.transaction.**deserialize**(*tx*)

cryptnox_cli.lib.cryptos.transaction.**serialize**(*txobj*, *include_witness=True*)

cryptnox_cli.lib.cryptos.transaction.**uahf_digest**(*txobj*, *i*)

cryptnox_cli.lib.cryptos.transaction.**signature_form**(*tx*, *i*, *script*, *hashcode=SIGHASH_ALL*)

cryptnox_cli.lib.cryptos.transaction.**der_encode_sig**(*v*, *r*, *s*)

cryptnox_cli.lib.cryptos.transaction.**der_decode_sig**(*sig*)

cryptnox_cli.lib.cryptos.transaction.**is_bip66**(*sig*)

Checks hex DER sig for BIP66 consistency

cryptnox_cli.lib.cryptos.transaction.**txhash**(*tx*, *hashcode=None*, *wtxid=True*)

cryptnox_cli.lib.cryptos.transaction.**public_txhash**(*tx*, *hashcode=None*)

cryptnox_cli.lib.cryptos.transaction.**bin_txhash**(*tx*, *hashcode=None*)

cryptnox_cli.lib.cryptos.transaction.**ecdsa_tx_sign**(*tx*, *priv*, *hashcode=SIGHASH_ALL*)

cryptnox_cli.lib.cryptos.transaction.**ecdsa_tx_verify**(*tx*, *sig*, *pub*, *hashcode=SIGHASH_ALL*)

cryptnox_cli.lib.cryptos.transaction.**ecdsa_tx_recover**(*tx*, *sig*, *hashcode=SIGHASH_ALL*)

cryptnox_cli.lib.cryptos.transaction.**mk_pubkey_script**(*addr*)

Used in converting p2pkh address to input or output script

cryptnox_cli.lib.cryptos.transaction.**mk_scripthash_script**(*addr*)

Used in converting p2sh address to output script

cryptnox_cli.lib.cryptos.transaction.**output_script_to_address**(*script*, *magicbyte=0*)

cryptnox_cli.lib.cryptos.transaction.**mk_p2w_scripthash_script**(*witver*, *witprog*)

Used in converting a decoded pay to witness script hash address to output script

cryptnox_cli.lib.cryptos.transaction.**mk_p2wpkh_redeemscript**(*pubkey*)

Used in converting public key to p2wpkh script

cryptnox_cli.lib.cryptos.transaction.**mk_p2wpkh_script**(*pubkey*)

Used in converting public key to p2wpkh script

cryptnox_cli.lib.cryptos.transaction.**mk_p2wpkh_scriptcode**(*pubkey*)

Used in signing for tx inputs

cryptnox_cli.lib.cryptos.transaction.**p2wpkh_nested_script**(*pubkey*)

cryptnox_cli.lib.cryptos.transaction.**deserialize_script**(*script*)

cryptnox_cli.lib.cryptos.transaction.**serialize_script_unit**(*unit*)

cryptnox_cli.lib.cryptos.transaction.**serialize_script**(*script*)

cryptnox_cli.lib.cryptos.transaction.**mk_multisig_script**(**args*)

**Parameters**

**args** – List of public keys to used to create multisig and M, the number of signatures required to spend

**Returns**

multisig script

cryptnox_cli.lib.cryptos.transaction.**verify_tx_input**(*tx*, *i*, *script*, *sig*, *pub*)

cryptnox_cli.lib.cryptos.transaction.**multisign**(*tx*, *i*, *script*, *pk*, *hashcode=SIGHASH_ALL*)

cryptnox_cli.lib.cryptos.transaction.**apply_multisignatures**(*txobj*, *i*, *script*, **args*)

cryptnox_cli.lib.cryptos.transaction.**is_inp**(*arg*)

cryptnox_cli.lib.cryptos.transaction.**select**(*unspent*, *value*)

### cryptnox_cli.lib.cryptos.wallet module

Simple wallet abstractions for addresses, balances, and signing flows.

This module is based on code from pybtctools: https://github.com/primal100/pybitcointools/blob/master/cryptos/wallet.py

**class** cryptnox_cli.lib.cryptos.wallet.**HDWallet**(*keystore*, *num_addresses=0*, *last_receiving_index=0*, *last_change_index=0*)

Bases: object

**__init__**(*keystore*, *num_addresses=0*, *last_receiving_index=0*, *last_change_index=0*)

**privkey**(*address*, *formt='wif_compressed'*, *password=None*)

**export_privkeys**(*password=None*)

**pubkey_receiving**(*index*)

**pubkey_change**(*index*)

**pubtoaddr**(*pubkey*)

**receiving_address**(*index*)

**change_address**(*index*)

**property receiving_addresses**

**property change_addresses**

**new_receiving_address_range**(*num*)

**new_change_address_range**(*num*)

**new_receiving_addresses**(*num=10*)

**new_change_addresses**(*num=10*)

**new_receiving_address**()

**new_change_address**()

**balance**()

**unspent**()

**select**()

**history**()

**sign**(*tx*, *password=None*)

**mksend**(*outs*)

**sign_message**(*message*, *address*, *password=None*)

**is_mine**(*address*)

**is_change**(*address*)

**account**(*address*, *password=None*)

**details**(*password=None*)

### cryptnox_cli.lib.cryptos.wallet_utils module

General wallet helper functions (formatting, hashing, conversions).

cryptnox_cli.lib.cryptos.wallet_utils.**hmac_sha_512**(*x*, *y*)

cryptnox_cli.lib.cryptos.wallet_utils.**rev_hex**(*s*)

cryptnox_cli.lib.cryptos.wallet_utils.**int_to_hex**(*i*, *length=1*)

**exception** cryptnox_cli.lib.cryptos.wallet_utils.**InvalidPassword**

Bases: Exception

**exception** cryptnox_cli.lib.cryptos.wallet_utils.**InvalidPasswordException**

Bases: Exception

**exception** cryptnox_cli.lib.cryptos.wallet_utils.**InvalidPadding**

Bases: Exception

cryptnox_cli.lib.cryptos.wallet_utils.**assert_bytes**(**args*)

porting helper, assert args type

cryptnox_cli.lib.cryptos.wallet_utils.**append_PKCS7_padding**(*data*)

cryptnox_cli.lib.cryptos.wallet_utils.**strip_PKCS7_padding**(*data*)

cryptnox_cli.lib.cryptos.wallet_utils.**aes_encrypt_with_iv**(*key*, *iv*, *data*)

cryptnox_cli.lib.cryptos.wallet_utils.**aes_decrypt_with_iv**(*key*, *iv*, *data*)

cryptnox_cli.lib.cryptos.wallet_utils.**EncodeAES**(*secret*, *s*)

cryptnox_cli.lib.cryptos.wallet_utils.**DecodeAES**(*secret*, *e*)

cryptnox_cli.lib.cryptos.wallet_utils.**pw_encode**(*s*, *password*)

cryptnox_cli.lib.cryptos.wallet_utils.**pw_decode**(*s*, *password*)

cryptnox_cli.lib.cryptos.wallet_utils.**is_new_seed**(*x*, *prefix=version.SEED_PREFIX*)

cryptnox_cli.lib.cryptos.wallet_utils.**seed_type**(*x*)

cryptnox_cli.lib.cryptos.wallet_utils.**is_seed**(*x*)

cryptnox_cli.lib.cryptos.wallet_utils.**inv_dict**(*d*)

cryptnox_cli.lib.cryptos.wallet_utils.**is_minikey**(*text*)

cryptnox_cli.lib.cryptos.wallet_utils.**minikey_to_private_key**(*text*)

cryptnox_cli.lib.cryptos.wallet_utils.**get_pubkeys_from_secret**(*secret*)

cryptnox_cli.lib.cryptos.wallet_utils.**xprv_header**(*xtype*)

cryptnox_cli.lib.cryptos.wallet_utils.**xpub_header**(*xtype*)

cryptnox_cli.lib.cryptos.wallet_utils.**number_of_significant_digits**(*number: float*) → int

### Module contents

Cryptocurrency utilities package - aggregates and re-exports submodules for easier imports.

The `cryptnox_cli.lib.cryptos` package provides cryptocurrency utilities and re-exports functionality from:

- `cryptnox_cli.lib.cryptos.blocks` - Block-related utilities
- `cryptnox_cli.lib.cryptos.composite` - Composite key handling
- `cryptnox_cli.lib.cryptos.deterministic` - Deterministic wallet functions
- `cryptnox_cli.lib.cryptos.main` - Main crypto operations
- `cryptnox_cli.lib.cryptos.mnemonic` - Mnemonic phrase handling
- `cryptnox_cli.lib.cryptos.specials` - Special cryptocurrency operations
- `cryptnox_cli.lib.cryptos.stealth` - Stealth address support
- `cryptnox_cli.lib.cryptos.transaction` - Transaction handling
- `cryptnox_cli.lib.cryptos.coins` - Coin implementations
- `cryptnox_cli.lib.cryptos.keystore` - Key storage utilities
- `cryptnox_cli.lib.cryptos.wallet` - Wallet functionality

All public members from these modules are available directly from `cryptnox_cli.lib.cryptos`.

### Module contents

The `cryptnox_cli.lib` package provides library functionality for cryptocurrency operations. The main subpackage is `cryptnox_cli.lib.cryptos` which contains comprehensive cryptocurrency utilities.

### cryptnox_cli.wallet package

### Subpackages

### cryptnox_cli.wallet.eth package

### Submodules

### cryptnox_cli.wallet.eth.api module

A basic Ethereum wallet library

`cryptnox_cli.wallet.eth.api.`**`address`**(*public_key: str*) → str

`cryptnox_cli.wallet.eth.api.`**`checksum_address`**(*public_key: str*) → str

**class** `cryptnox_cli.wallet.eth.api.`**`Api`**(*endpoint: str*, *network:* EthNetwork | *str*, *api_key: str*)

Bases: `object`

**`PATH = "m/44'/60'/0'/0/0"`**

**SYMBOL = 'eth'**

**__init__**(*endpoint: str*, *network:* EthNetwork | *str*, *api_key: str*)

**property block_number**

**contract**(*address=''*, *abi=''*)

**get_transaction_count**(*address: str*, *blocks: str = None*) → int

**get_balance**(*address: str*) → int

**property gas_price**

**property network**

**transaction_hash**(*transaction: Dict[str, Any]*, *vrs: bool = False*)

**push**(*transaction*, *signature*, *public_key*)

**class** cryptnox_cli.wallet.eth.api.**EthValidator**(*endpoint: str = 'publicnode'*, *network: str = 'sepolia'*, *price: int = 8*, *limit: int = 2500*, *derivation: str = 'DERIVE'*, *api_key=''*)

Bases: object

Class defining Ethereum validators

**__init__**(*endpoint: str = 'publicnode'*, *network: str = 'sepolia'*, *price: int = 8*, *limit: int = 2500*, *derivation: str = 'DERIVE'*, *api_key=''*)

**endpoint**
: Validator to validate endpoints for the Ethereum network

**network**
: Class for validating if value is part of the enum

**price**
: Class for validating if value is integer

**limit**
: Class for validating if value is integer

**derivation**
: Class for validating if value is part of the enum

**api_key**

**validate**()

### cryptnox_cli.wallet.eth.endpoint module

Module for endpoints that can be used for working with the Ethereum network

**class** cryptnox_cli.wallet.eth.endpoint.**EndpointValidator**(*valid_values: str = None*)

Bases: Validator

Validator to validate endpoints for the Ethereum network

**__init__**(*valid_values: str = None*)

**validate**(*value*)

**Parameters**
**value** – Evaluated value

**Returns**
None

**Raises**
**ValidationError** – Validation criteria not satisfied

**class** cryptnox_cli.wallet.eth.endpoint.**Endpoint**(*network:* EthNetwork, *api_key: str = ''*)

Bases: object

Abstract base class for interface for endpoint implementations

**__init__**(*network:* EthNetwork, *api_key: str = ''*)

**abstract property available_networks: List[EthNetwork]**
Ethereum networks handled by the endpoint implementation :rtype: List[enums.EthNetwork]

**Type**
return

**abstract property domain: str**
Domain that is used by the endpoint implementation :rtype: str

**Type**
return

**abstract property name: str**
Name of the endpoint implementation :rtype: str

**Type**
return

**abstract property provider: str**
Full URL that can be used as Web3 provider :rtype: str

**Type**
return

**class** cryptnox_cli.wallet.eth.endpoint.**InfuraEndpoint**(*network:* EthNetwork, *api_key: str = ''*)

Bases: Endpoint

Implementation of the Infura endpoint

**name = 'infura'**

**available_networks = ['MAINNET', 'KOVAN', 'GOERLI', 'RINKEBY', 'SEPOLIA']**

**__init__**(*network:* EthNetwork, *api_key: str = ''*)

**property domain: str**

Domain that is used by the endpoint implementation :rtype: str

**Type**
return

**property provider: str**

Full URL that can be used as Web3 provider :rtype: str

**Type**
return

**class** cryptnox_cli.wallet.eth.endpoint.**CryptnoxEndpoint**(*network:* EthNetwork, *api_key: str = ''*)

Bases: Endpoint

Implementation of the Cryptnox endpoint

**name = 'cryptnox'**

**available_networks = ['MAINNET', 'SEPOLIA']**

**property domain: str**

Domain that is used by the endpoint implementation :rtype: str

**Type**
return

**property provider: str**

Full URL that can be used as Web3 provider :rtype: str

**Type**
return

**class** cryptnox_cli.wallet.eth.endpoint.**PublicNodeEndpoint**(*network:* EthNetwork, *api_key: str = ''*)

Bases: Endpoint

Implementation of the PublicNode public RPC endpoint (free, no API key required)

**name = 'publicnode'**

**available_networks = ['MAINNET', 'SEPOLIA']**

**property domain: str**

Domain that is used by the endpoint implementation :rtype: str

**Type**
return

**property provider: str**

Full URL that can be used as Web3 provider :rtype: str

**Type**
return

**class** cryptnox_cli.wallet.eth.endpoint.**DirectEndpoint**(*url: str*, *network:* EthNetwork)

Bases: object

**__init__**(*url: str*, *network:* EthNetwork)

**property provider: str**

cryptnox_cli.wallet.eth.endpoint.**factory**(*endpoint: str*, *network:* EthNetwork, *api_key: str = ''*) → *Endpoint*

Factory method for Endpoint instances

**Parameters**

- **endpoint** – Name of the endpoint to use
- **network** – Ethereum network to use
- **api_key** – API key to use on the endpoint

**Type**
str

**Type**
*enums.EthNetwork*

**Type**
str

**Returns**
Return an Endpoint instance that can be used to get the urls

**Return type**
*Endpoint*

**Raises**
**ValueError** – In case endpoint with endpoint name wasn't found

### Module contents

Ethereum wallet utilities - aggregates and re-exports API components for easier imports.

The `cryptnox_cli.wallet.eth` package provides Ethereum wallet utilities and exports:

cryptnox_cli.wallet.eth.**address**(*public_key*)

Generate an Ethereum address from a public key. See `cryptnox_cli.wallet.eth.api.address()` for details.

**class** cryptnox_cli.wallet.eth.**Api**

Ethereum API interface. See `cryptnox_cli.wallet.eth.api.Api` for details.

cryptnox_cli.wallet.eth.**checksum_address**(*address_str*)

Convert an address to checksummed format. See `cryptnox_cli.wallet.eth.api.checksum_address()` for details.

**class** cryptnox_cli.wallet.eth.**EthValidator**

Ethereum address and transaction validator. See `cryptnox_cli.wallet.eth.api.EthValidator` for details.

### Submodules

### cryptnox_cli.wallet.btc module

A basic BTC wallet library

**class** cryptnox_cli.wallet.btc.**BtcNetworks**(**values*)

Bases: Enum

Class defining possible Bitcoin networks

**MAINNET = 'mainnet'**

**TESTNET = 'testnet'**

**TESTNET4 = 'testnet4'**

**class** cryptnox_cli.wallet.btc.**BlockCypherApi**

Bases: object

**__init__**(*api_key*, *network*)

**get_data**(*endpoint: str*, *params: Dict = None*, *data: bytes = None*) → None

**Return type**
None

**Parameters**

- **endpoint** – str
- **params** – dict
- **data** – bytes

**check_api_resp**() → None

**Return type**
None

**get_utx_os**(*addr: str*, *n_conf*) → List

**Parameters**

- **addr** – str
- **n_conf** – int (0 or 1)

**Returns**
List

**push_tx**(*tx_hex: str*) → Dict

**Parameters**
**tx_hex**

**Returns**
str

**get_key**(*key_char: str*) → Dict

:param key_char:str :return: Dict

**class** cryptnox_cli.wallet.btc.**BlkHubApi**

Bases: object

**__init__**(*network*, *api_key: str = ''*, *endpoint: str = ''*)

**static get_api**(*network: str*) → str

Get API url for given network

**Parameters**
**network** (str)

**Returns**
API url

**Return type**
str

**get_data**(*endpoint: str*, *params: Dict = None*, *data: bytes = None*) → None

**Parameters**

- **endpoint** – str
- **params** – dict
- **data** – bytes

**Returns**
None

**check_api_resp**() → None

**Return type**
None

**get_fee_estimates**(*blocks=6*) → int

**get_utx_os**(*addr: str*, *_n_conf: int*) → List

:param addr:str :param int _n_conf: 0 or 1 :return: list

**push_tx**(*tx_hex: str*) → List

**Parameters**
**tx_hex** – str

**Returns**
List

**get_key**(*key_char: str*) → List

**Parameters**
**key_char** – str

**Returns**
list

cryptnox_cli.wallet.btc.**test_addr**(*btc_addr: str*)

**Parameters**
**btc_addr** – str

**Returns**

**class** cryptnox_cli.wallet.btc.**BTCwallet**

Bases: object

**Parameters**

- **pubkey** – str
- **coin_type** – str
- **api**
- **connection**

**PATH = "m/44'/0'/0'/0/0"**

**__init__**(*pubkey: str*, *coin_type: str*, *api*, *card*) → None

**Parameters**

- **pubkey** – str
- **coin_type** – str
- **api**
- **connection**

**get_utx_os**(*n_conf: int = 0*)

**Parameters**
**n_conf** – int (0 or 1)

**Returns**

**get_balance**() → float

**Returns**
float

**get_fee_estimate**()

**prepare**(*to_addr: str*, *payment_value: float*, *fee: float*, *utx_os: List = None*) → float | int

**Parameters**

- **to_addr** – str
- **payment_value** – float
- **fee** – float
- **utx_os** – pre-fetched UTXOs (fetched in parallel with fee estimate)

**Returns**
Union[float, int]

**send**(*to_addr: str*, *payment_value: float*, *signature: List[bytes]*) → str

**Parameters**

- **to_addr** – str
- **payment_value** – float

**Returns**
str

**static balance_fm_utxos**(*utxos*) → float

**Parameters**
**utxos**

**Returns**
float

**static select_utxos**(*amount: float*, *utxos*) → List

**Parameters**

- **amount** – float
- **utxos**

**Returns**
float

**class** cryptnox_cli.wallet.btc.**BtcValidator**(*network: str = 'testnet'*, *fees: int = 2000*, *derivation: str = 'DERIVE'*, *api_key: str = ''*, *endpoint: str = ''*)

Bases: object

Class defining Bitcoin validators

**__init__**(*network: str = 'testnet'*, *fees: int = 2000*, *derivation: str = 'DERIVE'*, *api_key: str = ''*, *endpoint: str = ''*)

**network**
Class for validating if value is part of the enum

**fees**
Class for validating if value is integer

**derivation**
Class for validating if value is part of the enum

**api_key**

**endpoint**

### cryptnox_cli.wallet.validators module

Module for defining validators

**exception** cryptnox_cli.wallet.validators.**ValidationError**
Bases: Exception

Exception for indicating that validation criteria are not met

**class** cryptnox_cli.wallet.validators.**Validator**(*valid_values: str = None*)
Bases: ABC

Validator class defining setters and getters

**__init__**(*valid_values: str = None*)

**abstractmethod validate**(*value*)

**Parameters**
**value** – Evaluated value

**Returns**
None

**Raises**
**ValidationError** – Validation criteria not satisfied

**class** cryptnox_cli.wallet.validators.**AnyValidator**(*valid_values: str = None*)

Bases: Validator

**validate**(*value*)

**Parameters**
**value** – Evaluated value

**Returns**
None

**Raises**
**ValidationError** – Validation criteria not satisfied

**class** cryptnox_cli.wallet.validators.**IntValidator**(*max_value: int = None*, *min_value: int = None*)

Bases: Validator

Class for validating if value is integer

**__init__**(*max_value: int = None*, *min_value: int = None*)

**validate**(*value*) → int

**Parameters**
**value** – Evaluated value

**Returns**
None

**Raises**
**ValidationError** – Validation criteria not satisfied

**class** cryptnox_cli.wallet.validators.**EnumValidator**(*current_enum*)

Bases: Validator

Class for validating if value is part of the enum

**__init__**(*current_enum*)

**validate**(*value*) → None

**Parameters**
**value** – Evaluated value

**Returns**
None

**Raises**
`ValidationError` – Validation criteria not satisfied

**class** cryptnox_cli.wallet.validators.**UrlValidator**(*valid_values: str = None*)

Bases: `Validator`

Class for validating URLs

**validate**(*value: str*) → str

**Parameters**
**value** – Evaluated value

**Returns**
None

**Raises**
`ValidationError` – Validation criteria not satisfied

cryptnox_cli.wallet.validators.**is_int**(*value: Any*) → bool

### Module contents

The `cryptnox_cli.wallet` package provides wallet functionality for various cryptocurrencies. It includes Bitcoin wallet operations, Ethereum wallet utilities, and validation functions.

## 3.1.2 Submodules

## 3.1.3 cryptnox_cli.config module

## 3.1.4 cryptnox_cli.enums module

Module for defining enumerations used throughout the CryptnoxPro application.

**class** cryptnox_cli.enums.**EthNetwork**(**values*)

Bases: `Enum`

Class defining possible Ethereum networks

**MAINNET = 1**

**KOVAN = 42**

**GOERLI = 5**

**RINKEBY = 4**

**SEPOLIA = 11155111**

**class** cryptnox_cli.enums.**SolanaNetwork**(**values*)

Bases: `Enum`

Class defining possible Solana networks

**MAINNET = 1**

**DEVNET = 2**

**TESTNET = 3**

**class** cryptnox_cli.enums.**Command**(**values*)

Bases: Enum

**BTC = 'btc'**

**CARD_CONFIGURATION = 'card_conf'**

**CHANGE_PIN = 'change_pin'**

**CHANGE_PUK = 'change_puk'**

**CONFIG = 'config'**

**ETH = 'eth'**

**HISTORY = 'history'**

**INFO = 'info'**

**INITIALIZE = 'init'**

**CARD = 'list'**

**SERVER = 'server'**

**RESET = 'reset'**

**SEED = 'seed'**

**SOLANA = 'solana'**

**UNLOCK_PIN = 'unlock_pin'**

**USER_KEY = 'user_key'**

**TRANSFER = 'transfer'**

**CERTIFICATE = 'cert'**

### 3.1.5 cryptnox_cli.constants module

Centralized constants for the Cryptnox CLI.

### 3.1.6 cryptnox_cli.interactive_cli module

### 3.1.7 cryptnox_cli.main module

### 3.1.8 Module contents

The main cryptnox_cli package (Cryptnox CLI) exports:

cryptnox_cli.**__version__: str = "1.0.5"**

Current version of the Cryptnox CLI package.

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